

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2- Concept Mapping

Course Code:	CSA0307	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	N. Punith Kumar	Reg. No.:	192525156
Section / Batch:	2025	Date:	13/7/26

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Punith Kumar.

Instructions

- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

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(13/7)

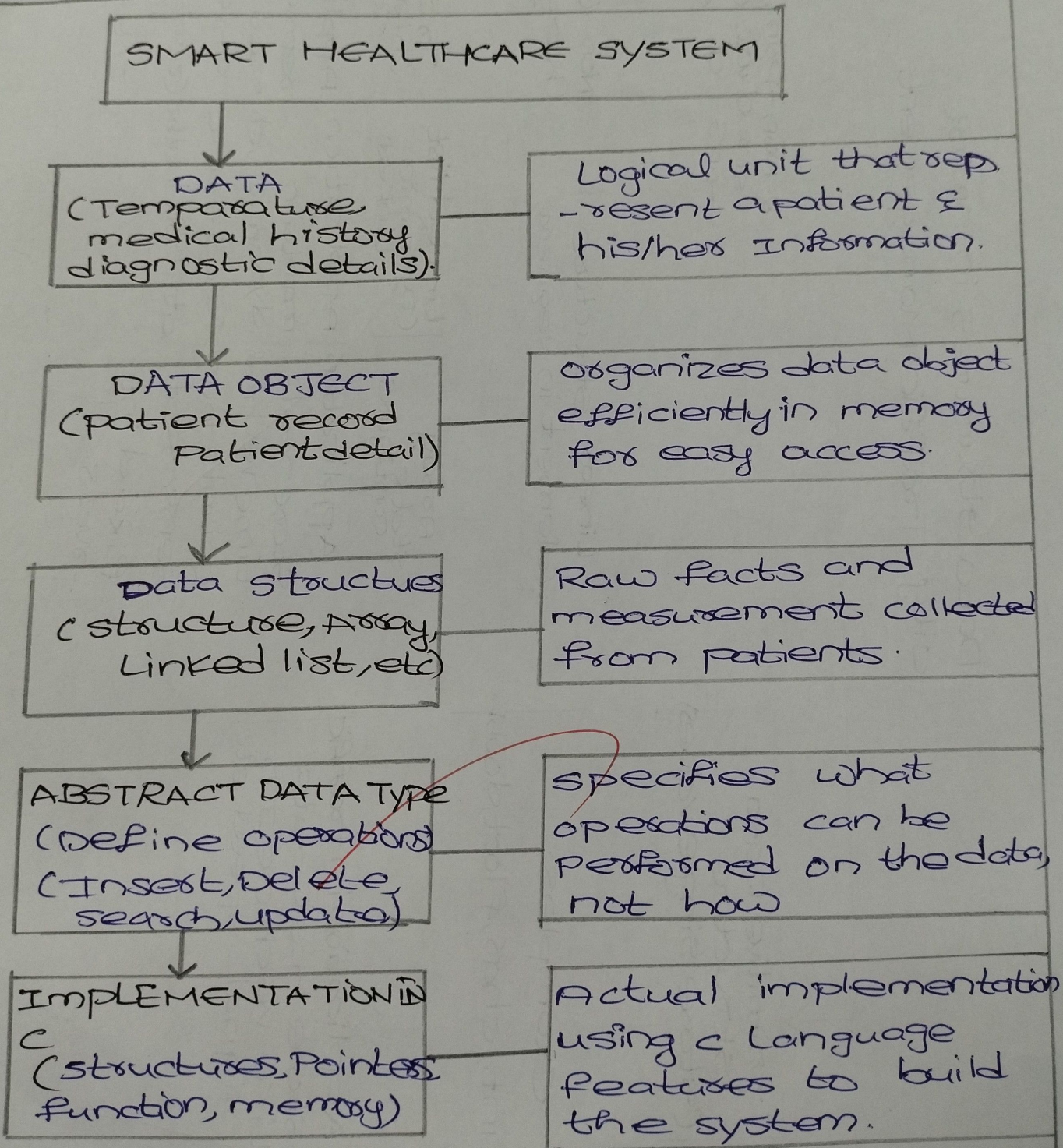
Assessment - concept mapping

Name :- N. punith kumar

Reg NO :- 192535156

COURSE
Name &
code : CSA0307 - Data Structure.

I)



Data structures in smart city Traffic Management

1. Primitive data structure
(store single values)

Examples:-

int, char, float, double

Application in traffic
store basic data
like vehicle ID,
signal, speed etc

2. Non-primitive data structure (store collection of values)

Linear structure
(element in sequence)

Array
(static contiguous)

Application

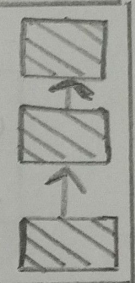
store real-time vehicle counts from sensors at fixed intervals.



Linked list
(Dynamic Linked)

Application

maintain dynamic vehicle queues at toll booths

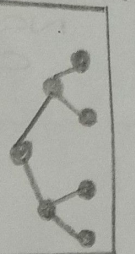


Non-Linear structure
(hierarchical or network relationships)

Tree
(hierarchical)

Application

represent road network hierarchy (zones, sectors, roads for intersection)



Graph.
(network based)

Application

model road network as intersections and roads for optional route finding

